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Earth Embeddings as Products

Taxonomy, Ecosystem, and Standardized Access

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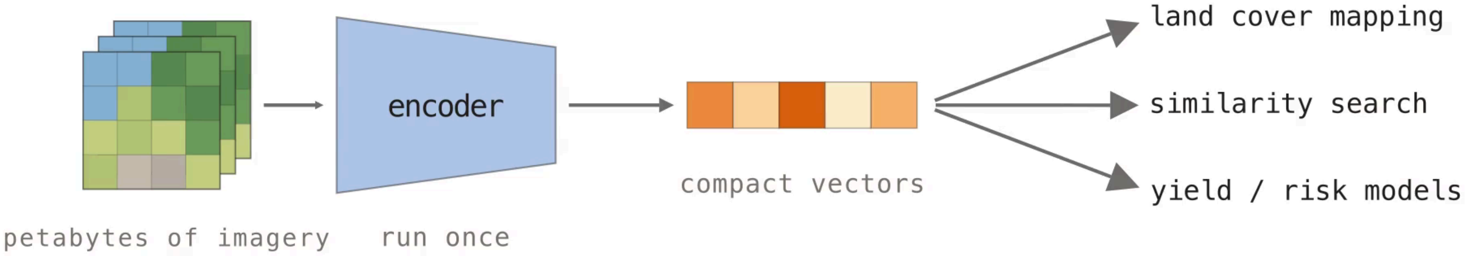
Embedding products move model inference out of the user's workflow

Public EO archives are already petabyte-scale. NASA's EOSDIS alone holds **178.7 PB** and grows by 160 TB per day. Foundation models can summarize this data, but every user re-pays the preprocessing and GPU bill. Pre-computed embedding products run the model once and ship the vectors as reusable data.



NASA ESDS annual metrics, FY2025 · Bommasani et al., 2021 — foundation model framing.

SCOPE

A foundation model is not an embedding product

Foundation models (dynamic inference)

These are distributed as public weights (DOFA, OlmoEarth). Users manage preprocessing and GPU inference themselves. This is flexible but raises a hardware and engineering barrier.

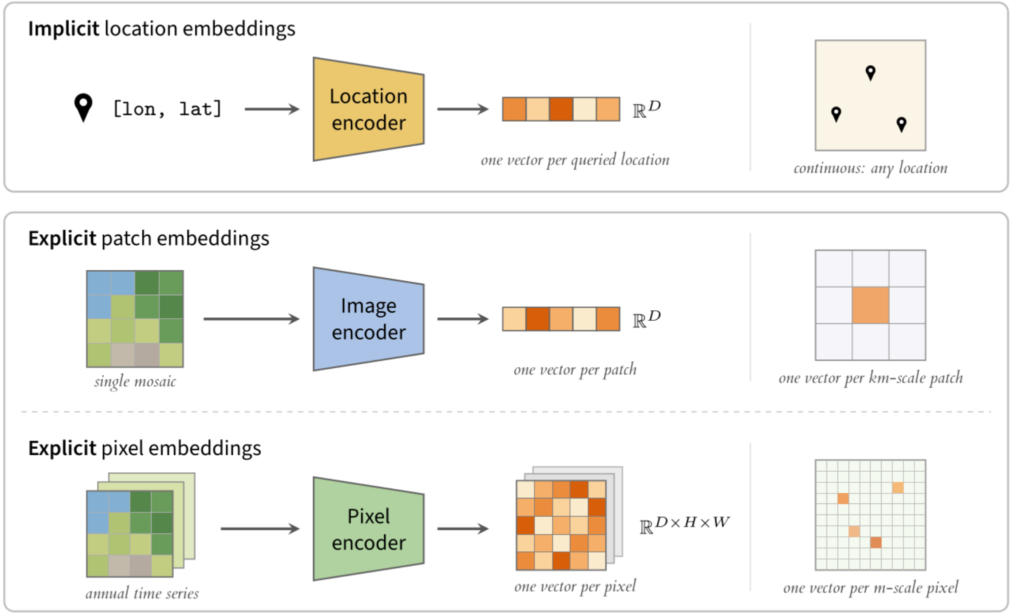
Embedding products (static data)

These are distributed as pre-computed vector archives (Google Satellite Embedding). The assets are frozen and versioned, so analysis proceeds without the model or its compute.

A product is pinned to the data snapshot it was computed on, so **treating a product as a proxy for its model invites invalid generalization claims**.

Fang, Stewart, Corley, Zhu, Azizpour — Earth Embeddings as Products, IGARSS 2026, §II.

Three families: location, patch, and pixel embeddings



Stewart, Fang, Corley, Zhu — Earth Embeddings (book chapter, 2026), Fig. 1; figure design adapted from Klemmer et al., 2025.

Patch products summarize km-scale tiles for retrieval

All known patch embedding products as of July 2026. *sparse spatial or temporal coverage.

Product	Extent	Resolution	Years	Dim.	Dtype
MOSAIKS (2021–25)	USA / Global	1 km / 0.01°	2018 / 2019	8192 / 4000	float32
Clay v0 / v1.5 (2024–26)	USA / Global	154 m – 5.12 km	2010–2025*	768–1024	float32
Major TOM (2024–26)	Global*	120 m – 3.84 km	2016–2024*	256–2048	float32
Earth Index (2025)	Global	320 m	2024	384	float32
Copernicus-Embed (2025)	Global	0.25°	2021	768	float32

Major TOM is the largest family, with more than a dozen model variants on one shared grid. Every product stores float32. One vector summarizes an entire mosaic tile, which suits retrieval more than dense mapping.

Earth Embeddings (book chapter, 2026), Table 2 — abridged to family level.

Pixel products store one vector per pixel for each year

All known pixel embedding products as of July 2026. Every product is built from annual time series. *sparse coverage.

Product	Extent	Resolution	Years	Dim.	Dtype
Presto Embeddings (2025)	Togo	10 m	2019–2020	128	uint16
Tessera Embeddings (2025)	Global*	10 m	2017–2025*	128	int8 → float32
Google Satellite Embedding (2025)	Global	10 m	2017–2025	64	int8 → float64
Embedded Seamless Data (2026)	Global	30 m	2000–2024	12	uint16 → float32

Google Satellite Embedding covers the full Sentinel era at 64 dimensions. **Embedded Seamless Data** trades resolution for 25 years of Landsat history. Storage pressure drives the int8 and uint16 dtypes.

Earth Embeddings (book chapter, 2026), Table 3.

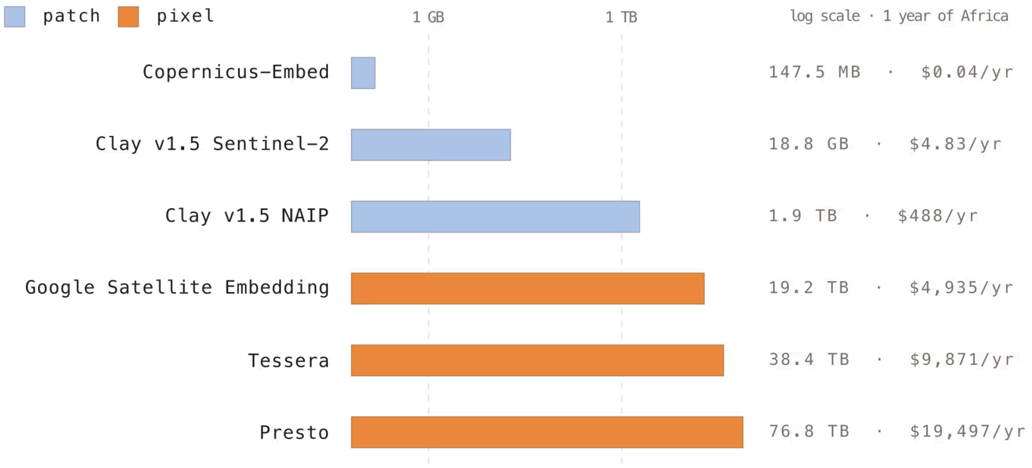
Each product ships with its own formats, grids, and hosting

- Products are scattered across **Source Cooperative** (Clay, Earth Index), Hugging Face (Major TOM, Copernicus-Embed), Google Earth Engine (Google, Presto), and private university servers (Tessera).
- Formats range from **GeoParquet** to GeoTIFF with implicit CRS assumptions to raw NumPy arrays with no CRS or bounds ("numbers and a prayer").
- Each product defines its own **tiling grid** (Major TOM grid, MGRS, custom), so cross-product comparison starts with reprojection.
- **One flipped coordinate** in the Google Satellite Embedding rasters required patches to **GDAL, rasterio, and TorchGeo** before standard tools could read them.

Every team solves distribution independently. The integration tax is paid once per product, per user.

Corley — The Technical Debt of Earth Embedding Products, cloudnativegeo.org, Feb 2026.

Storage for one continent-year spans five orders of magnitude



Patch products stay in the MB–GB range, while dense 10 m pixel products reach tens of TB. A single full download of Presto's Africa coverage adds ~**$6.9k in egress** on top of storage.

Earth Embeddings (book chapter, 2026), Table 8 — AWS S3 Standard, us-west-2 · egress from Corley, 2026.

Hosting and format choices decide whether a product is usable

- **Hugging Face** enforces repo storage caps and API rate limits, so bulk pulls of TB-scale products throttle or fail.
- **Tessera** ships raw `.npy` tiles with coordinates and metadata in a sidecar file. NumPy supports no HTTP range requests, so reads pull whole tiles that a **COG or Zarr would stream** — GeoTIFF re-implemented without its streaming or metadata.
- **Google Satellite Embedding** was reachable only inside Earth Engine until **Taylor Geospatial rehosted it on Source Cooperative**, which adds free egress, an HTTP cross-region proxy, and CORS so browsers can stream it.
- Several products are so **sparse in space and time** that no common footprint exists for comparing them.

Corley — The Technical Debt of Earth Embedding Products, cloudnativegeo.org, Feb 2026 · Google Satellite Embedding rehost: source.coop, 2026.

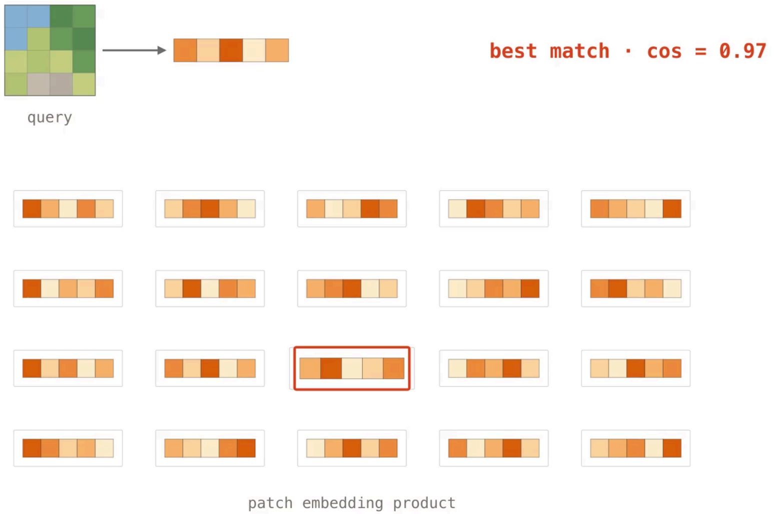
Patch products are reproducible, but pixel models are often proprietary

- **Clay, Earth Index, and Copernicus-Embed** release the full pipeline (code, weights, data, embeddings) under permissive licenses.
- **Major TOM**'s CC-BY-SA pretraining data makes every derived embedding copyleft, which deters commercial users.
- **AlphaEarth Foundations and ESDNet** keep code and weights proprietary. The embeddings are CC-BY, but no one outside can regenerate or audit them.
- **No product ships checksummed data.** Upstream archives keep reprocessing imagery, so the exact training and inference inputs are already unrecoverable.

Earth Embeddings (book chapter, 2026), Tables 4–6 — license provenance from data to weights to embeddings.

TorchGeo treats embeddings as first-class geospatial datasets

- TorchGeo has loaders for **every known embedding product**, plus the generating models and weights. **Presto and Tessera** were added upstream for this work.
- Reprojection, rasterization, spatiotemporal intersection, and sampling are built in.
- These workflows used to require **four or more repositories** and custom loaders. With TorchGeo each is about 20 lines of code.



Stewart et al., 2022 — TorchGeo · Fang et al., IGARSS 2026, §IV.

Search and land cover mapping in TorchGeo

Search and retrieval: query by example

```
# torchgeo.datasets / torchgeo.models
earthindex = EarthIndexEmbeddings('data/ei')
s2 = Sentinel2('data/s2')
image = s2[xmin:xmax, ymin:ymax]['image'] / 10_000

model = vit_small_patch14_dinov2(
    ViTSmall14_DINOv2_Weights.SENTINEL2_ALL_SOFTCON)
embed = model(image)

cos = CosineSimilarity(dim=0)
for sample in iter(earthindex):
    sim = cos(embed, sample['embedding'])
    ...  # keep the argmax, plot the match
```

Land cover mapping: crop types across Europe

```
# torchgeo.datasets / torchgeo.samplers
tessera = TesseraEmbeddings('data/tessera')
eurocrops = EuroCrops('data/ec', download=True)
dataset = tessera & eurocrops  # intersection

train_roi = box(-10, 35, 10, 60)  # West EU
test_roi  = box( 10, 35, 30, 60)  # East EU
train_ds, test_ds = roi_split(
    dataset, [train_roi, test_roi])

# random patches → fit a k-NN / linear probe
# gridded patches → stitch a map of Europe
```

Earth Embeddings (book chapter, 2026), §4 Listings 1–2 — abridged; geographic train/test splits are built in.

Embeddings improve mapping tasks but degrade under spatial transfer

Task	Embeddings	Finding
Cropland mapping, Togo	Presto, Google	Presto + Random Forest gives the best F1
Tree species, Dutch forest inventory	Presto, Google, Tessera	+2–9 points over hand-designed time-series features
Landslide susceptibility, TW·HK·IT	Google	64-d embeddings beat conventional conditioning factors
Poverty mapping, Sub-Saharan Africa	Google + GNN	large storage/preprocessing savings vs. raw Sentinel-2
Scene classification, EuroSAT-Embed	Google, Tessera, OlmoEarth	pooling choice cuts the geographic gap by >50%
Fusion across six tasks	Google, Tessera, GeoCLIP, SatCLIP	fused embeddings beat the best single model in 4 of 6

Performance drops under **spatial transfer**, and annual composites wash out sub-annual dynamics. Validate with geographic splits and simple baselines before trusting any product.

Zvonkov et al. · Ishikawa et al., 2025 · Cheng et al., 2026 · Pettersson & Daoud, 2025 · Corley et al., 2026 · van der Plas et al., 2026 · Ma et al., 2026.

Recommended formats, provenance, and compression for new products

Embedding type	Format
Location / implicit	ONNX or PyTorch
Patch / region	GeoParquet
Pixel, snapshot	GeoTIFF or GeoZarr
Pixel, time series	GeoZarr

Ship a model card with sensors, temporal window, CRS and grid, dtype, quantization transform, and license. Store the metadata in the files, not the docs.

- **int8 quantization** has negligible accuracy cost. Google and Tessera already ship int8.
- **PCA to 64 dimensions with int8** gives 64× compression with <2% accuracy loss.
- **Binary quantization** adds another 32× and still recovers ~65% of float32 nearest neighbors, which is enough for candidate retrieval.
- Ship a **runnable benchmark**, not a private leaderboard.

Earth Embeddings (book chapter, 2026), §7 · Corley & Robinson, 2026 — TerraBit compression study.

TAKEAWAYS

Embeddings are usable today; standards are the remaining work

Pick the family by task: implicit for location context, patch for retrieval, pixel for dense mapping. Validate under geographic splits against simple baselines. Open problems are ocean and atmosphere coverage, uncertainty layers, and shared benchmarks (identical models differ by more than ten points across papers).

Chapter Earth Embeddings — arXiv, August 2026

Survey github.com/hfangcat/Awesome-Geospatial-Embeddings

Code github.com/torchgeo/torchgeo

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